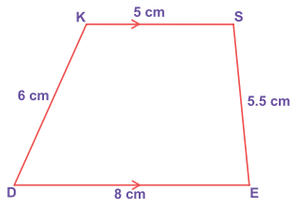
5.11.4 Constructing a trapezium when its four sides are given

Example 5.29

Construct a trapezium DESK in which $\overline{DE}$ is parallel to $\overline{KS}$, DE=8*cm*, ES=5.5*cm*, KS =5*cm* and KD=6*cm*. Find also its area.

Solution:

Given:
DE=8*cm*, ES=5.5*cm*, KS =5*cm*, KD=6*cm* and $\overline{DE} \parallel \overline{KS}$



Rough diagram

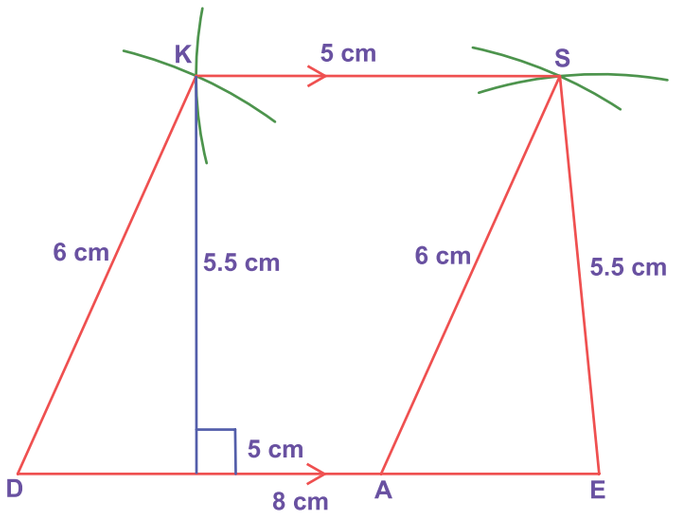


Fig. 5.48

Steps:

1. Draw a line segment DE= 8*cm*.
2. Mark the point A on DE such that DA=5*cm*.
3. With A and E as centres, draw arcs of radii 6*cm* and 5.5*cm* respectively. Let them cut at S. Join AS and ES.
4. With D and S as centres, draw arcs of radii 6*cm* and 5*cm* respectively. Let them cut at K. Join DK and KS.
5. DESK is the required trapezium.

Calculation of Area:

Area of the trapezium DESK $= \frac{1}{2} \times h \times (a+b)\, sq.units$

$= \frac{1}{2} \times 5.5 \times (8+5) = 35.75\, sq.cm$

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